

DHENUKA CHELUMALLA

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Master's student in Computer Science at the University of Georgia with a strong foundation in Python, SQL, and cloud-native data engineering. Microsoft Certified: Fabric Data Engineer Associate and Databricks Certified Data Engineer Associate, with hands-on expertise in building end-to-end ETL pipelines, RESTful APIs, and distributed data processing using Apache Spark and PySpark across Azure and AWS. Skilled in CI/CD, data validation, and pipeline monitoring to deliver reliable, production-ready data solutions.

Education

University of Georgia

Aug. 2025 – May. 2027

Master of Science in Computer Science

Athens, Georgia

- **Courses:** Algorithms, Computer Networks, Advanced Cloud Computing, Software Engineering, Distributed Computing Systems, Optimization in AI

Technical Skills

Languages: Python, SQL, JavaScript, HTML, CSS

Azure: ADF, Synapse, ADLS, Azure Monitor, Microsoft Fabric

AWS: S3, EC2, Redshift, RDS, Lambda

Data Engineering: Databricks, Snowflake, Apache Airflow, Spark, PySpark, ETL, Data Warehousing, Data Modeling

Databases: MySQL, PostgreSQL, MongoDB, Amazon RDS

Machine Learning: PyTorch, CNNs, NumPy, pandas, Datasets (MNIST, CIFAR-10, ImageNet)

Tools: Docker, Jenkins, Git, VS Code, PyCharm, IntelliJ, Jira, CI/CD

Visualization: Power BI, Matplotlib

Experience

Graduate Research Assistant

Aug. 2026 – Present

University of Georgia

Athens, Georgia

- Researching **neural operator surrogate models** for pore-scale reactive transport (velocity and concentration fields in rock pore structures) under the **DOE Genesis Project**.
- Extending the **PRT-DeepONet** model from **2D to 3D** in **PyTorch** using **3D CNNs**.
- Building and validating **3D data pipelines** for pore geometries, velocity and concentration fields, including spatial sampling and normalization.
- Running numerical simulations and model training on the **GACRC HPC** cluster using **Slurm**.
- Evaluating **generalization to unseen pore geometries** through geometry-level train/test splits and dataset integrity checks.

Data Engineer Intern

Aug. 2024 – May 2025

Intern

Remote

- Built **end-to-end ETL pipelines** with **Python (Pandas)** and **Azure Data Factory**, handling data **extraction, validation, transformation**, and **loading** into structured databases (**Azure SQL Database**).
- Designed and implemented data processing logic using **Python** and **SQL** on **Microsoft Azure**, improving **data processing speed** and **pipeline reliability**.
- Analyzed complex datasets and **optimized queries** to support **reporting needs**, downstream analytics, and **performance improvements**.
- Developed and integrated **RESTful APIs** to streamline access to processed data and enable communication between data services.
- Built and maintained **CI/CD pipelines** using **Azure DevOps** to automate **build, testing**, and **deployment**, and set up **monitoring** and **alerting** with **Azure Monitor** and **Application Insights** to ensure pipeline health and **data quality**.
- Collaborated with the engineering team using **Git** for **version control**, **code reviews**, and **feature-based branching**.

Certifications

Microsoft Certified: Fabric Data Engineer Associate | [Link](#)

Databricks Certified Data Engineer Associate | [Link](#)

Docker Foundations Professional Certificate | [Link](#)

Generative AI | [Link](#)

Artificial Intelligence Fundamentals | [Link](#)

SQL (Advanced) | [Link](#)

Projects

Review Intelligence: Aspect-Based Review Analysis | *PyTorch, HuggingFace, Ollama(Phi-3), Streamlit* Jun 2026

- Built an **end-to-end NLP pipeline** (streaming ingestion, cleaning, **Parquet** store, reproducible orchestration) that turns **45K+** Amazon reviews into per-aspect sentiment, grounded summaries, and fake-review trust scores, with **90%** data retention.
- Fine-tuned **DistilBERT** for aspect-level sentiment on 16K weakly-labeled reviews, improving **macro-F1** by **+0.20** ($0.48 \rightarrow 0.69$) over a lexicon baseline through fine-tuning and decision-threshold calibration.
- Engineered a hallucination-proof summary layer using a local **Phi-3** LLM with citation verification, rejecting any claim not grounded in a real review snippet (**100% grounded** on evaluated products).
- Shipped an interactive **Streamlit** app and a reproducible, config-driven, unit-tested repo running entirely on free local models (no paid APIs).

Intelligent Container Orchestration Platform | *Docker, FastAPI, Redis, Prometheus, Grafana* Apr 2026

- Built a self-healing microservices platform simulating cloud-native infrastructure on a single machine with **Docker**, featuring four **FastAPI** services communicating through an **API gateway** and **Redis pub/sub**.
- Developed a custom **Python orchestration engine** that monitors container health and resource usage in real time, automatically restarting failed containers within **2 seconds** and logging every recovery to a Redis audit trail.
- Exposed **FastAPI** endpoints for service registration, health checks, and metrics, enabling **dynamic service discovery** and event-driven coordination across the cluster.
- Implemented **auto-scaling** recommendations from live CPU/memory metrics and full observability via **Prometheus** and **Grafana** dashboards tracking request rates, p50/p95 latency, and self-healing events, secured with stateless **JWT authentication**.

OptiReduce: Tail-Optimal Gradient Aggregation for Distributed ML | *NumPy, Sockets (TCP/UDP)* Mar 2026

- Implemented and benchmarked three distributed gradient-aggregation strategies (**Parameter Server**, **Ring AllReduce**, **OptiReduce**) on a four-node cluster using the Python socket API with runtime-selectable **TCP/UDP** transport and custom chunking/reassembly for large tensors.
- Built the OptiReduce design with **UDP-based bounded transport**, adaptive timeouts from RTT percentiles, and **Hadamard-transform** encoding for resilience to dropped gradients, plus a coordinator service for peer discovery.
- Developed a **noise-injection module** simulating packet loss, straggler delays, and CPU contention to evaluate robustness under realistic cloud conditions.
- Demonstrated OptiReduce maintained a stable **0.99×** communication ratio under stragglers (vs. $1.64\times$ PS, $1.86\times$ Ring) and cut tail latency to a **1.1–1.2×** Pmax/P50 ratio.

CineBook: Movie Ticket Booking Platform | *Next.js, NestJS, Firebase (Firestore), JWT* May 2025

- Built a full-stack movie ticket booking app with a **Next.js** frontend (shadcn/ui) for browsing movies, selecting seats, and booking tickets, backed by a **NestJS** REST API.
- Structured the backend into modular **NestJS** controllers and services with dependency injection, improving maintainability and testability of the booking workflow.
- Designed **Firebase Firestore** data models for users, movies, and bookings, and secured the API with stateless **JWT authentication** and cookie-based sessions.
- Implemented **email notifications** (Nodemailer) and **scheduled jobs** (node-cron) for booking confirmations and automated cleanup, improving reliability and user experience.

Image Classification using CNN on CIFAR-10 Dataset | *Python, TensorFlow, PyTorch* Jan 2025

- Built a Convolutional Neural Network (CNN) to classify 60,000 CIFAR-10 images into 10 categories including airplane, cat, dog, and truck.
- Conducted data preprocessing, model architecture design, training, evaluation, and prediction visualization.
- Applied **data augmentation**, **dropout**, and **batch normalization** to reduce overfitting and improve generalization across the ten image categories.
- Achieved 78% training accuracy and 71% validation accuracy, strengthening expertise in deep learning, computer vision, and model optimization techniques.